ASTROPHYSICS LABORATORY

Organization — goals, deadlines

Team

Introduction

Lecture notes for the LA2025@IST, Part I, by A. Mourão, J. D. Duarte, R. P Santos

TÉCNICO LISBOA

centra
30th Anniversary

Part I: Observational Astrophysics

Prof. Ana Mourão, Dr. João Duarte, Dra Rita Santos

Part II: Stellar Astrophysics

Prof. Ilídio Lopes, Dr. Diogo Campelo

Observational Astrophysics

Hands-on sessions: scientific observing projects

Goal:
develop a scientific project to be elaborated in small groups in which students go through the different scientific steps:

- writing an observing proposal
- planning and performing observations with a telescope
- reducing and analyzing the data
- writing a research report

Evaluation

Observing Proposal: 15%	11 set 17:00
Observation and observing logs: 20%	Data of observation
Data products (images, figures, codes)	10 October
Final short report/paper: 45%	10 October 23:59
Final presentation: 20%	15 October

Observational Astrophysics

PROJECTS

1) Stars
2) stellar clusters
3) Planetary nebula
4) you name it ...

WEBSITE:
https://fenix.tecnico.ulisboa.pt/disciplinas/LA112/2025-2026/1-semestre

FORUM:
Use it to ask questions and discuss concepts, problems, etc.

TÉCNICO LISBOA

Bibliography

BOOKS:

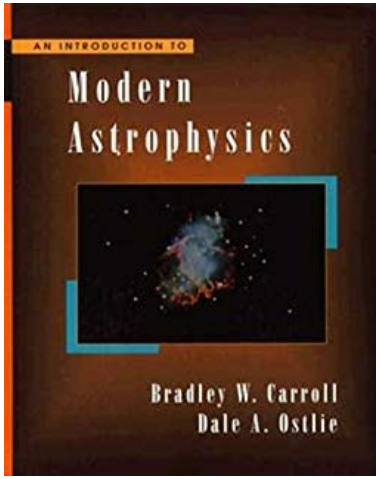
B.Carroll and D. Ostlie: An Introduction to Modern Astrophysics, 2nd Edition, 2013
M. Zeilik and S. A. Gregory, A Pratical Guide to CCD Astronomy, Cambridge University Press, 1998

Sites:

http://spiff.rit.edu/classes/phys445/syllabus.html
http://spiff.rit.edu/classes/phys445
http://burro.case.edu/Academics/Astr306/
http://www.astronomynotes.com
https://web.njit.edu/~gary/202

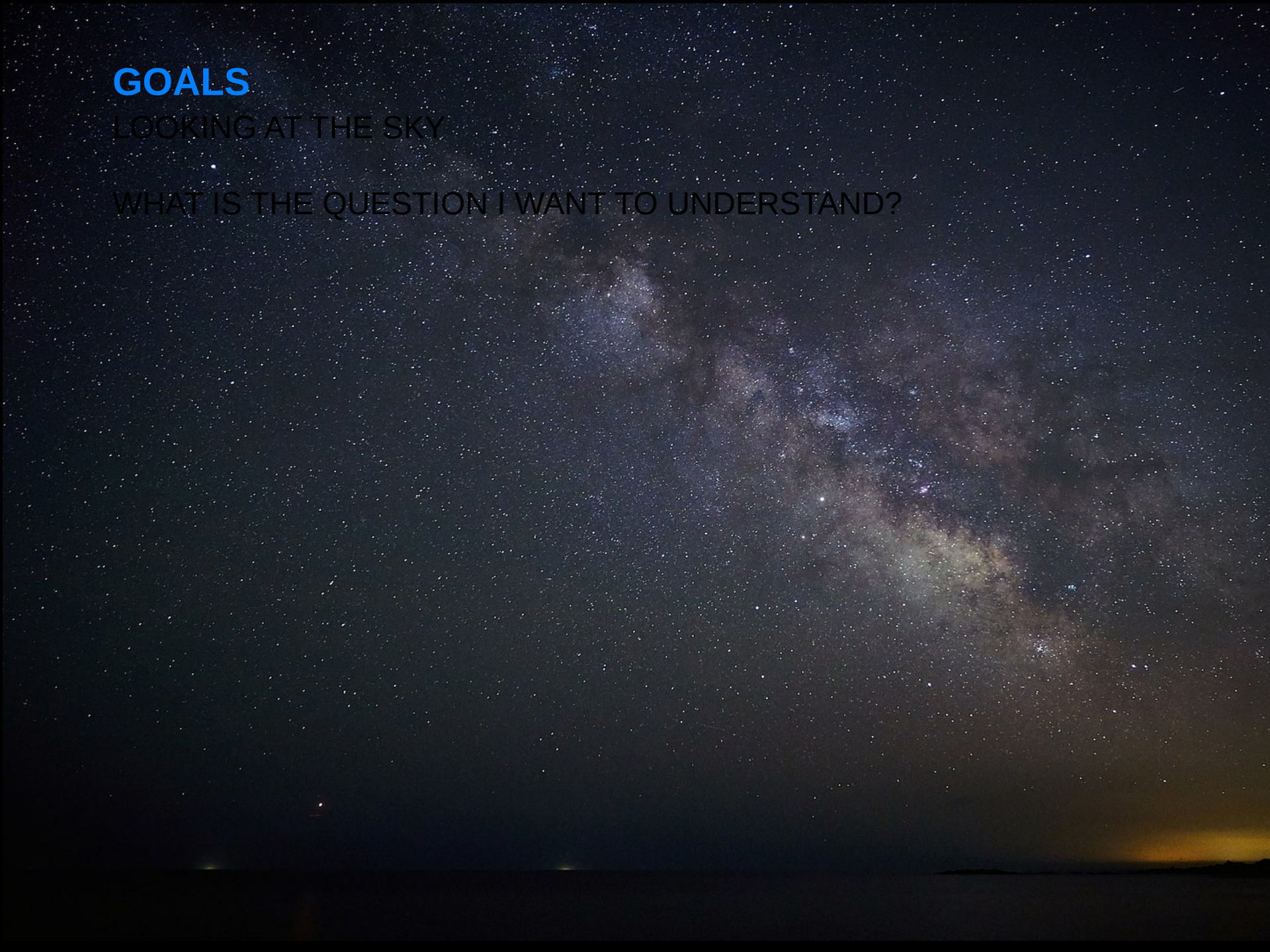
Sites with working packages and tools:

http.//www.airmass.org
See AL2025 website: here



OBSERVATIONAL ASTROPHYSICS @ AL 2025




GOALS
LOOKING AT THE SKY
WHAT IS THE QUESTION I WANT TO UNDERSTAND?





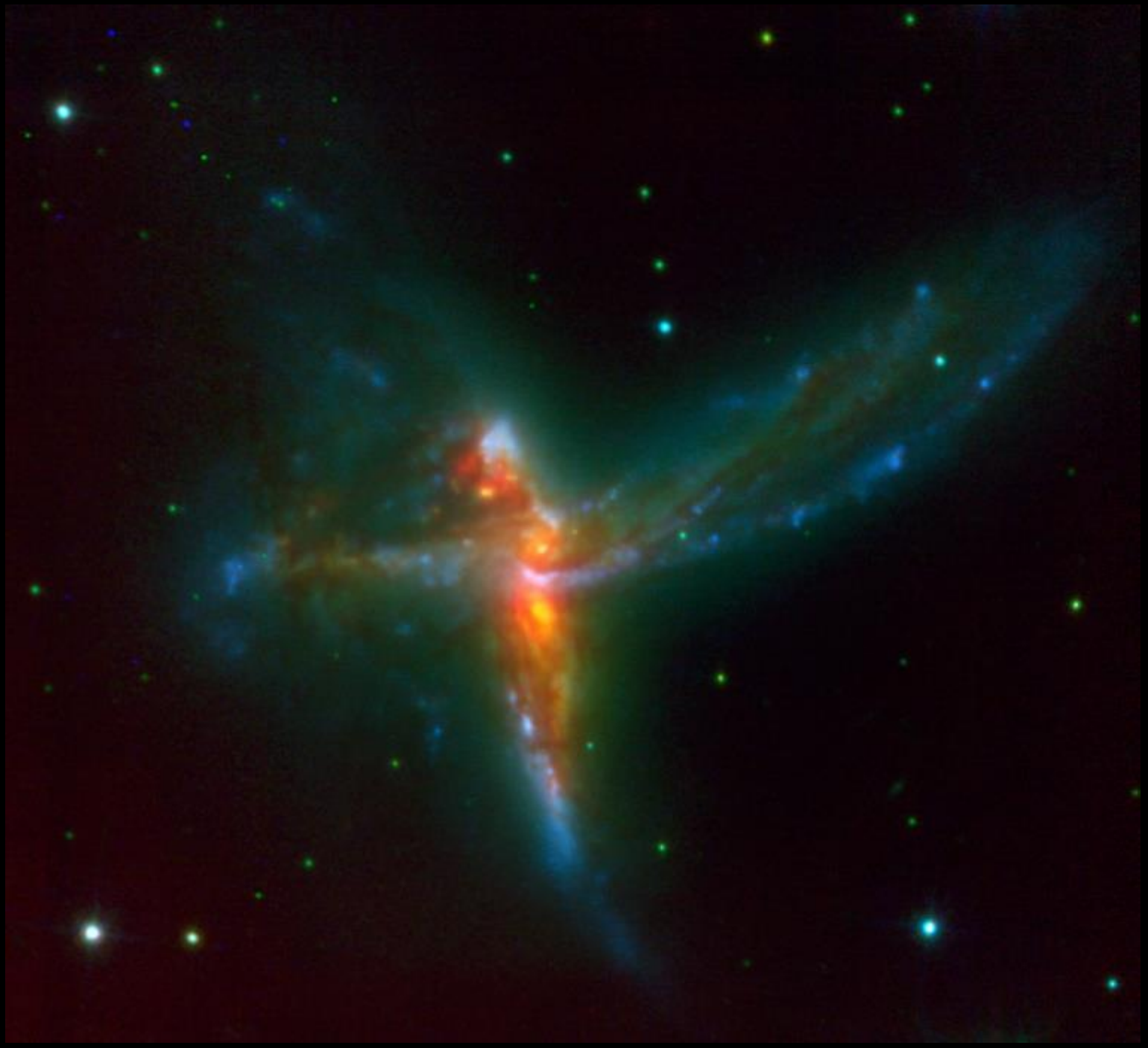




Star cluster Westerlund
1500ly by ESO



Abel cluster
credit ESO



'Bird' interacting galax
6500Mly by ESO+HST

ORIGIN AND EVOLUTION

Earth

Moon

Planets

Comets

Stars

Galaxies

Solar system

Stellar Clusters

Exo planets

Clusters of Galaxies

UNIVERSE

SCALES

Earth	$R=6{,}4\ 10^{6}\ m$
Sun-Earth distance	$D=1AU=1{,}5\ 10^{11}\ m$
Sun-Neptune distance	$D=30AU$
Diameter of the Milky Way	$D=150\ kly=150\ 10^{16} m$
Cluster of galaxies	$\sim 10^{23} m$
Atomic size	$\sim 10^{-10} m$

ASTROPHYSICS

WE PROBE THE UNIVERSE

AT DIFFERENT AGES

AT DIFFERENT SCALES

THE LAWS OF THE UNIVERSE ARE

THE LAWS OF PHYSICS

Explain what is seen in the sky

THE UNIVERSE IS A LABORATORY



OBSERVATIONAL ASTRONOMY

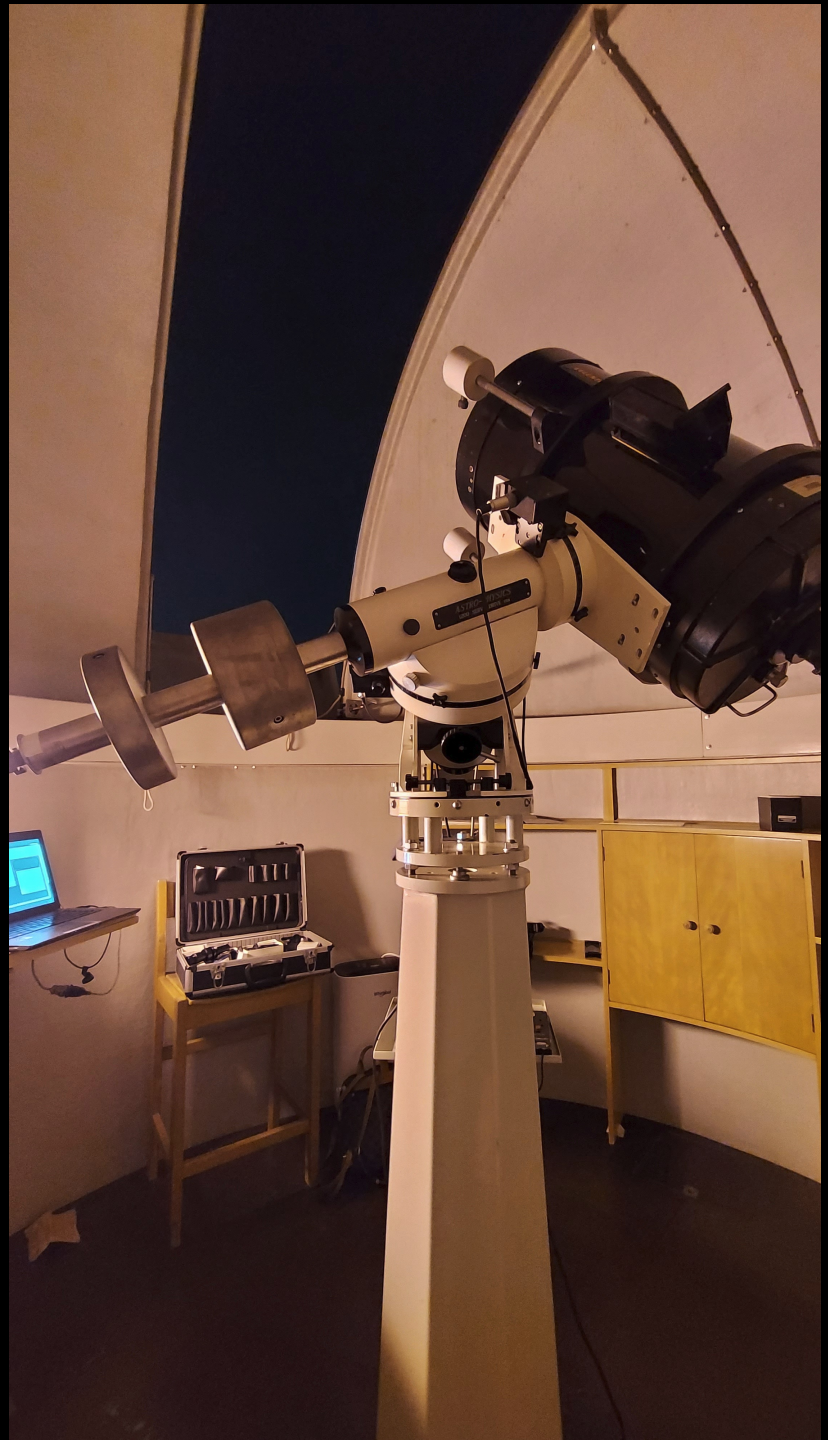

OBSERVATIONAL ASTRONOMY
THE UNIVERSE TO DISCOVER





Galileu
1609


www.atacamaphoto.com




MEGAPRIME,CFHTLS, 340 M



CANON 12 M


SAMSUNG
14:18
Wed, Feb 26
Tap to add city
Phone
Contacts
Internet
Apps
SAMSUNG
14:18 Wed, February 26
PorSer
Por ser de Movistar, en talleres Bos..
13:18
Clear
Emergency call
Unlock via Google
Alternative password
Movistar
ANDROID AUT





Can we really

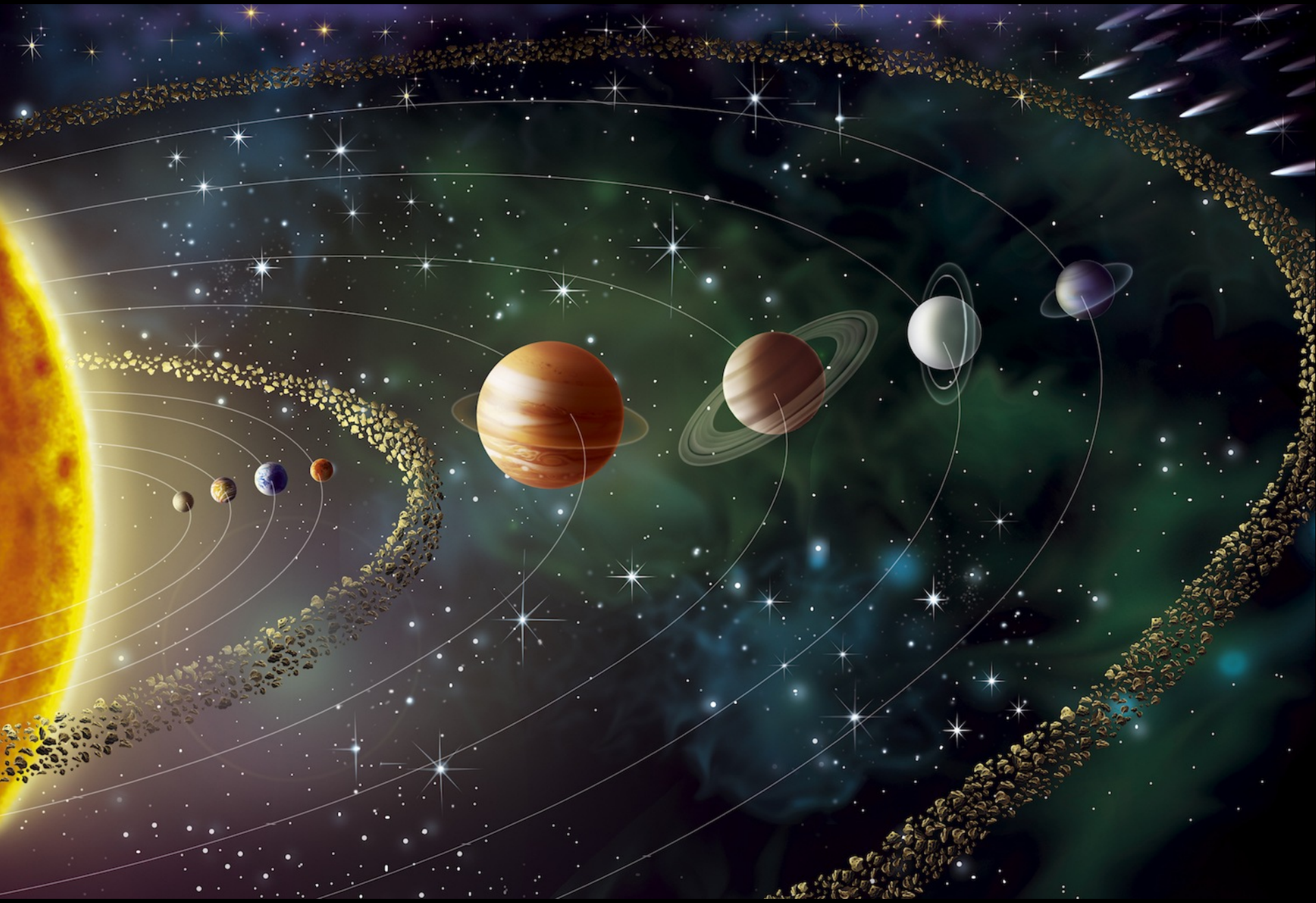
understand the Universe?

FUNDAMENTAL INTERACTIONS

GRAVITATION

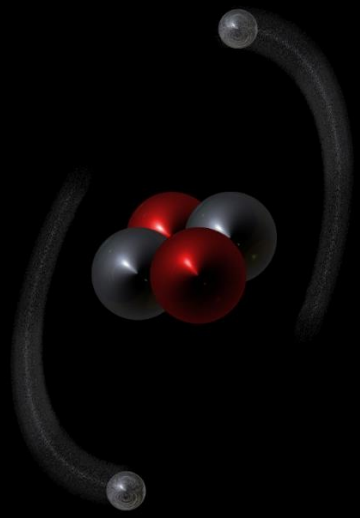


Newton 1642-1727

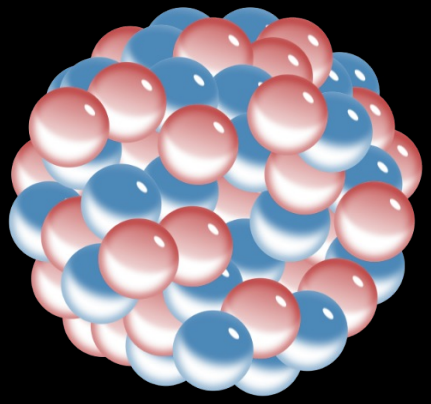


AAS

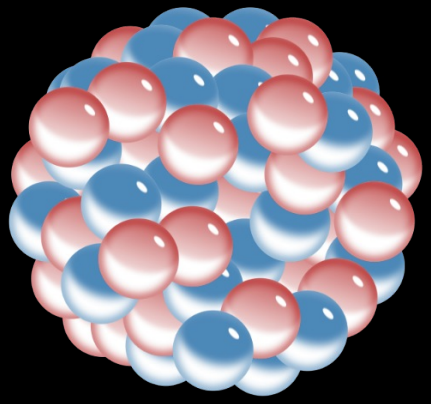
ELECTROMAGNETIC INTERACTION

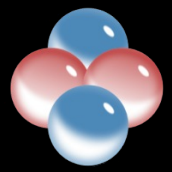


STRONG INTERACTION



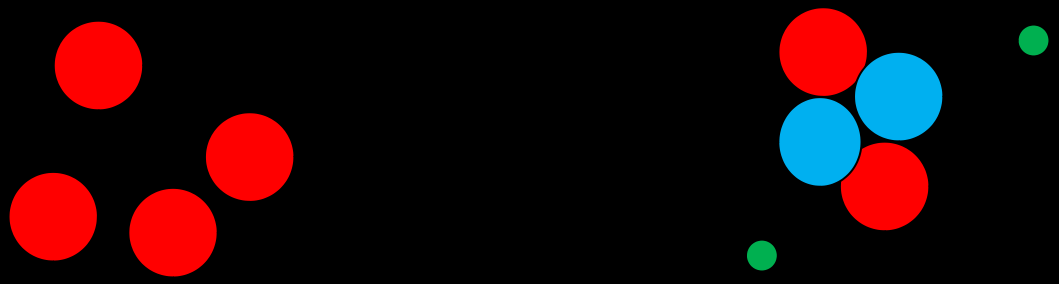
WEAK INTERACTION





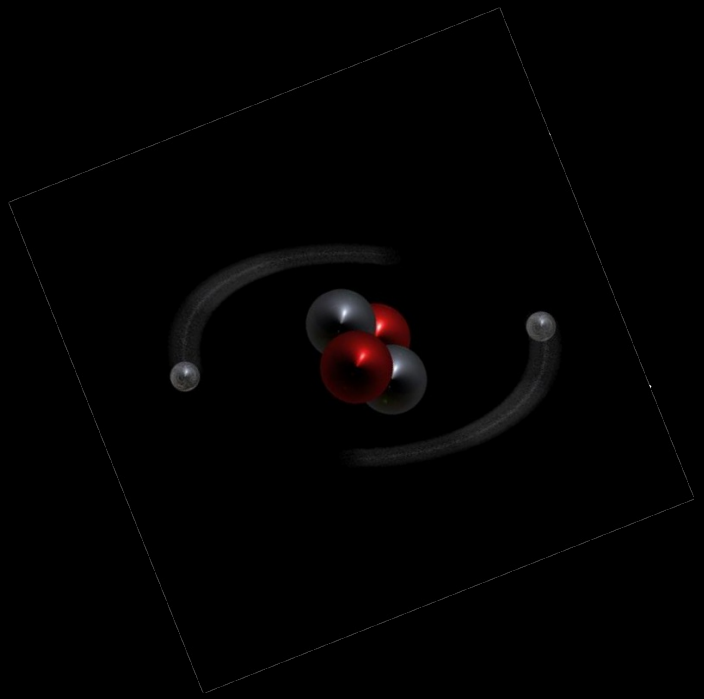
STRONG AND ELECTROWEAK INTERACTIONS

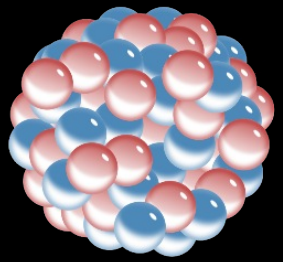
EXPLAIN THE SOURCE OF ENERGY IN STARS

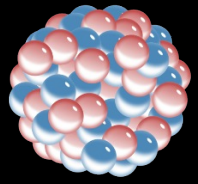


$$4\,p \longrightarrow He + 2\,\nu + 2e^{+} + \text{Energy}$$















radius

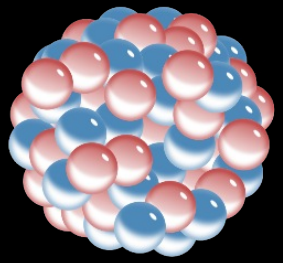
temperature

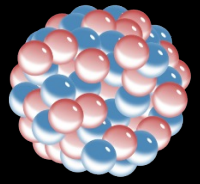
evolution

mass

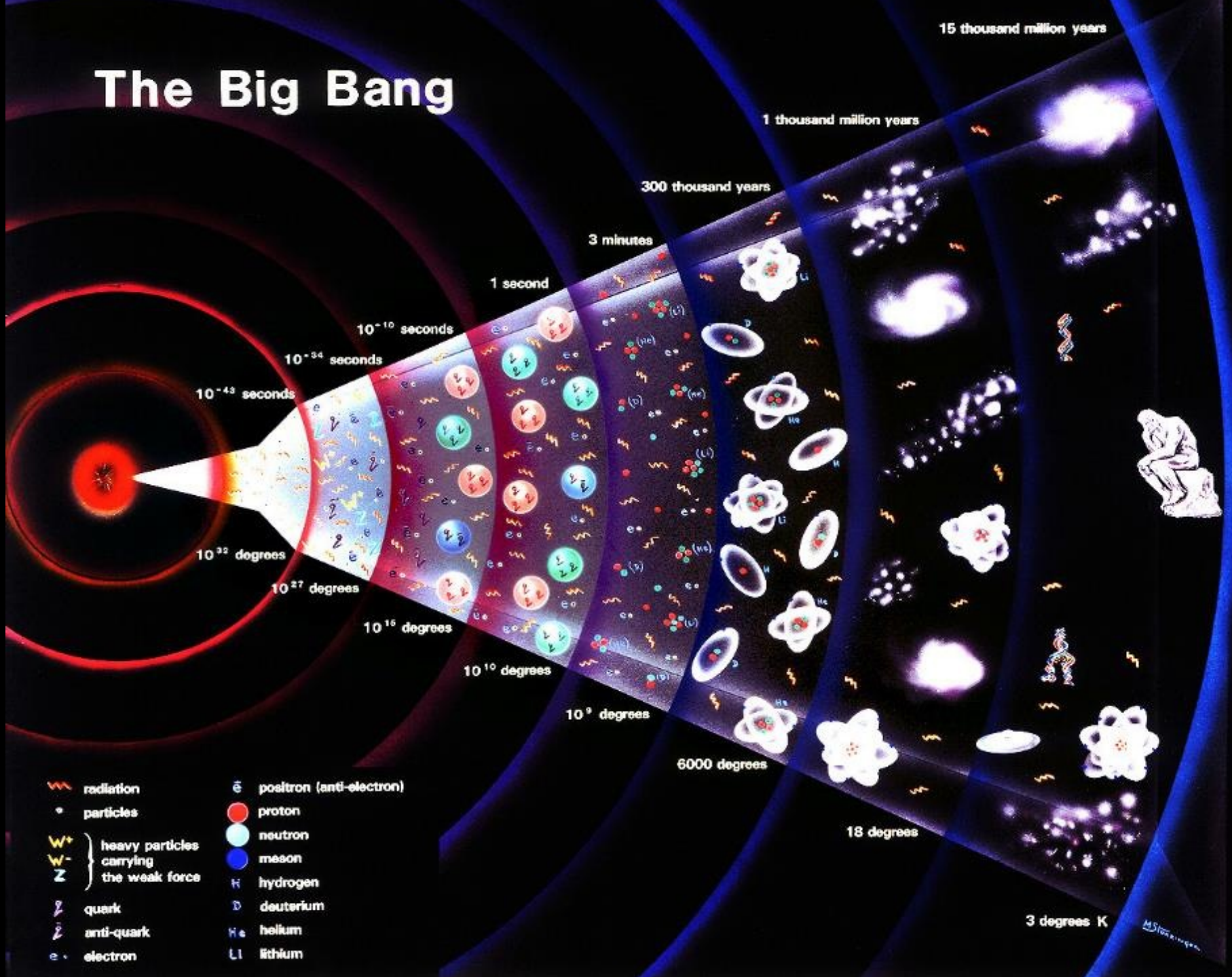
luminosity

colapse








The Big Bang
15 thousand million years
1 thousand million years
300 thousand years
3 minutes
1 second
10^{-10} seconds
10^{-34} seconds
10^{-43} seconds
10^{32} degrees
10^{27} degrees
10^{15} degrees
10^{10} degrees
10^{9} degrees
6000 degrees
18 degrees
3 degrees K
radiation
particles
W+
W-
Z
heavy particles carrying the weak force
quark
anti-quark
electron
positron (anti-electron)
proton
neutron
meson
H hydrogen
D deuterium
He helium
Li lithium

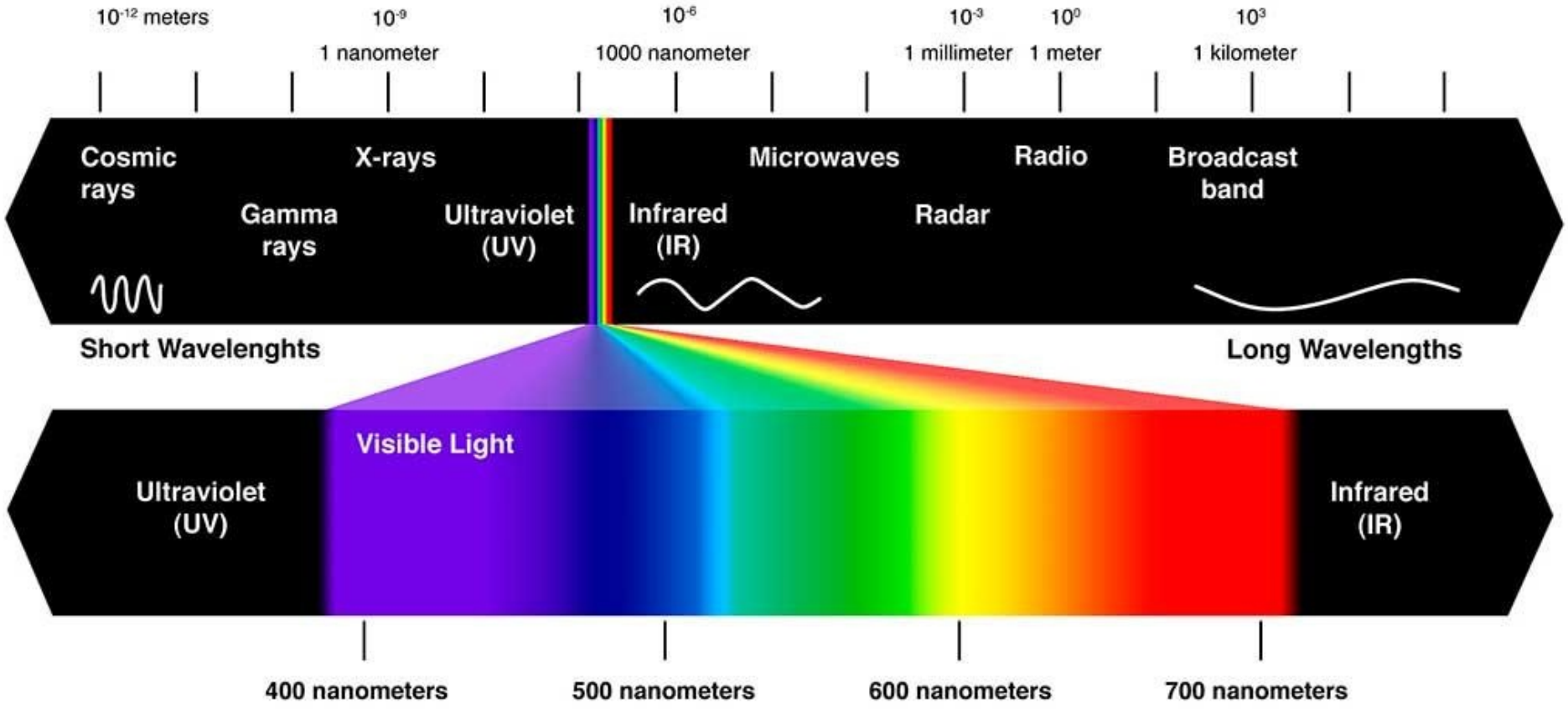
OBSERVATIONAL ASTROPHYSICS @ AL 2024

What are we looking for?

- What?
- Where?
- When?
- Why?
- How?

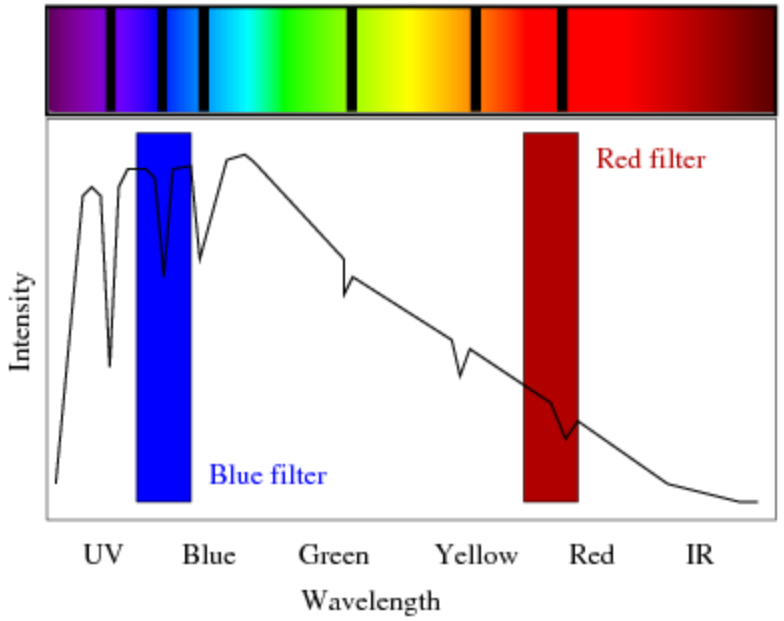


LIGHT FROM THE SOURCE

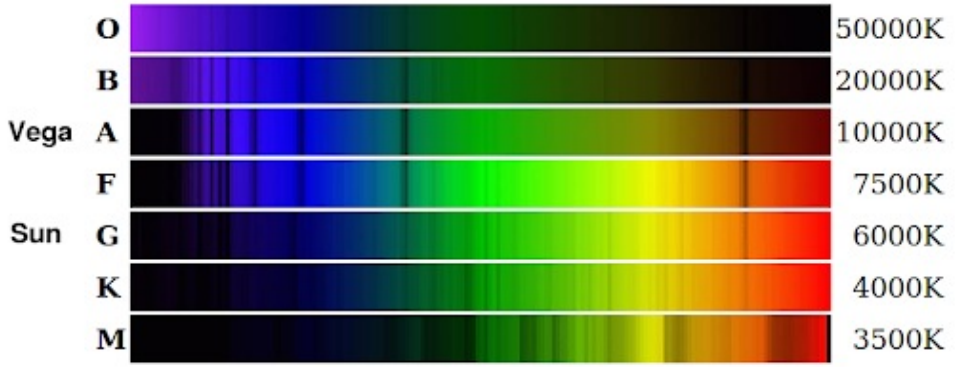


OBSERVATIONS

PHOTOMETRY



SPECTROSCOPY



HOW TO CHOOSE THE WAVELENGTH?

PROPERTIES OF THE OBJECT

DISTANCE

AVAILABLE TELESCOPES

Accelerated expansion of the Universe

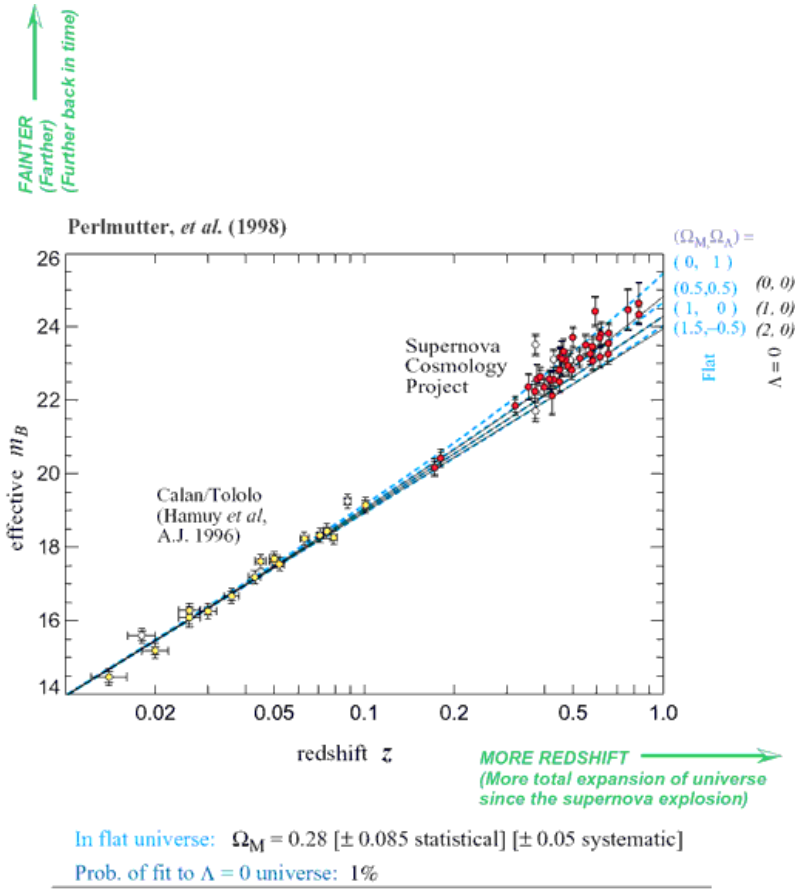


Saul Perlmutter, SCP



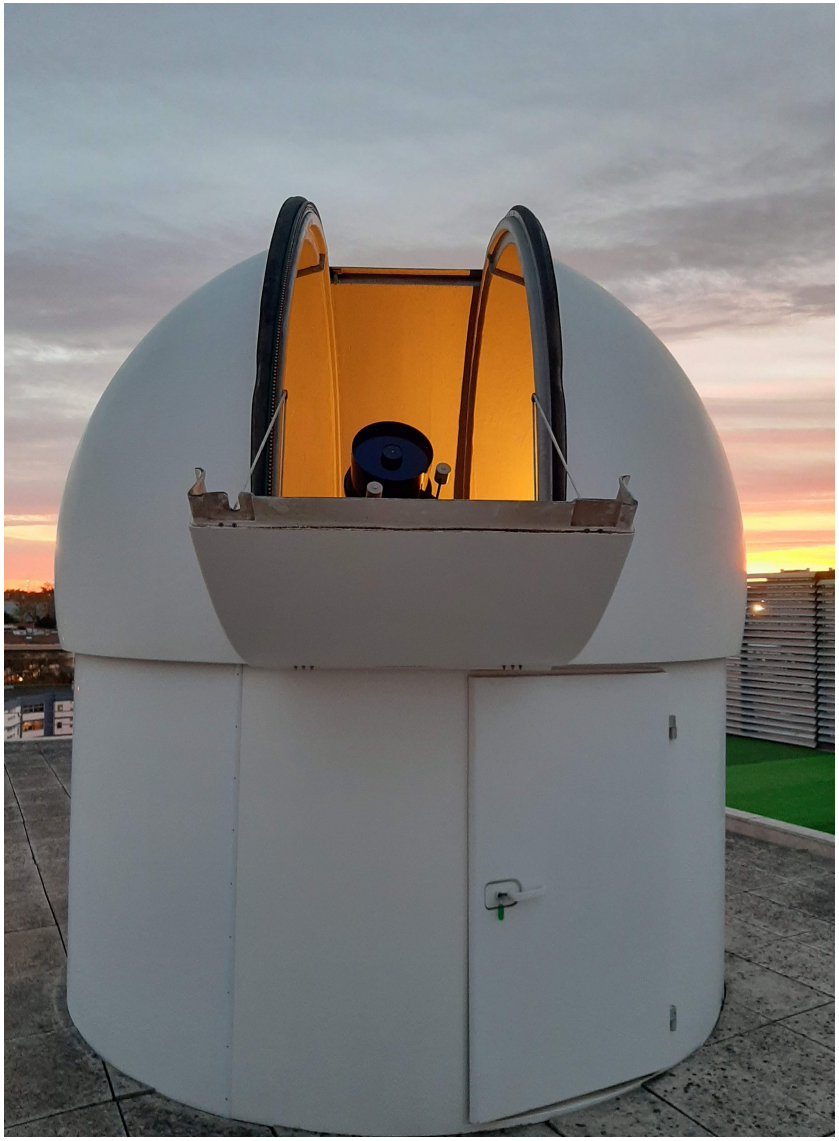


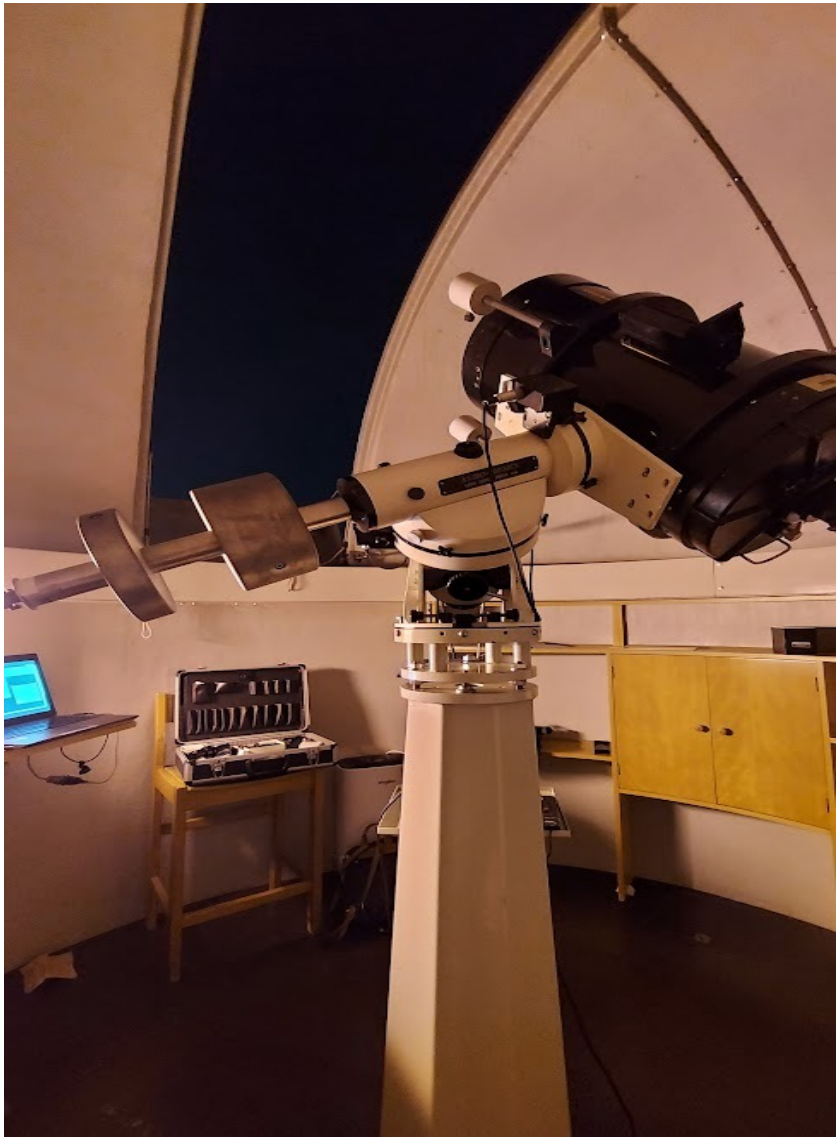
Adam Riess e Brian Schmidt, HZT



Nobel prize 2011

OBSERVATIONAL ASTROPHYSICS @ AL 2025





ASTRO-PHYSICS
1200 SERVO DRIVE

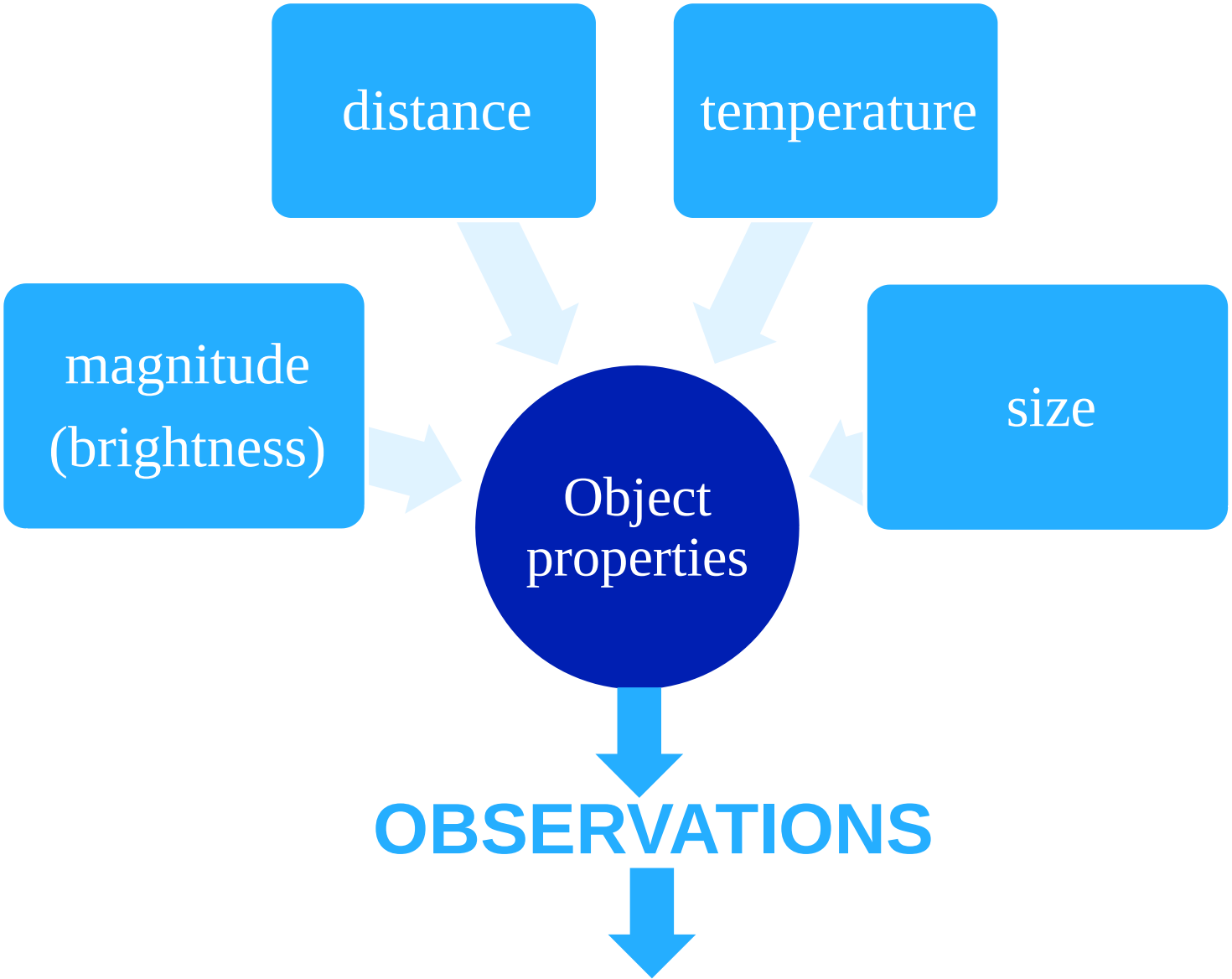
OBSERVATIONAL ASTROPHYSICS @ AL 2024



TARGETS=OBJECTS IN OUR SKY



OBSERVATIONAL ASTROPHYSICS @ AL 2025







D=7mm

$D_{Telescope}$=14”= 356mm

R= 2580

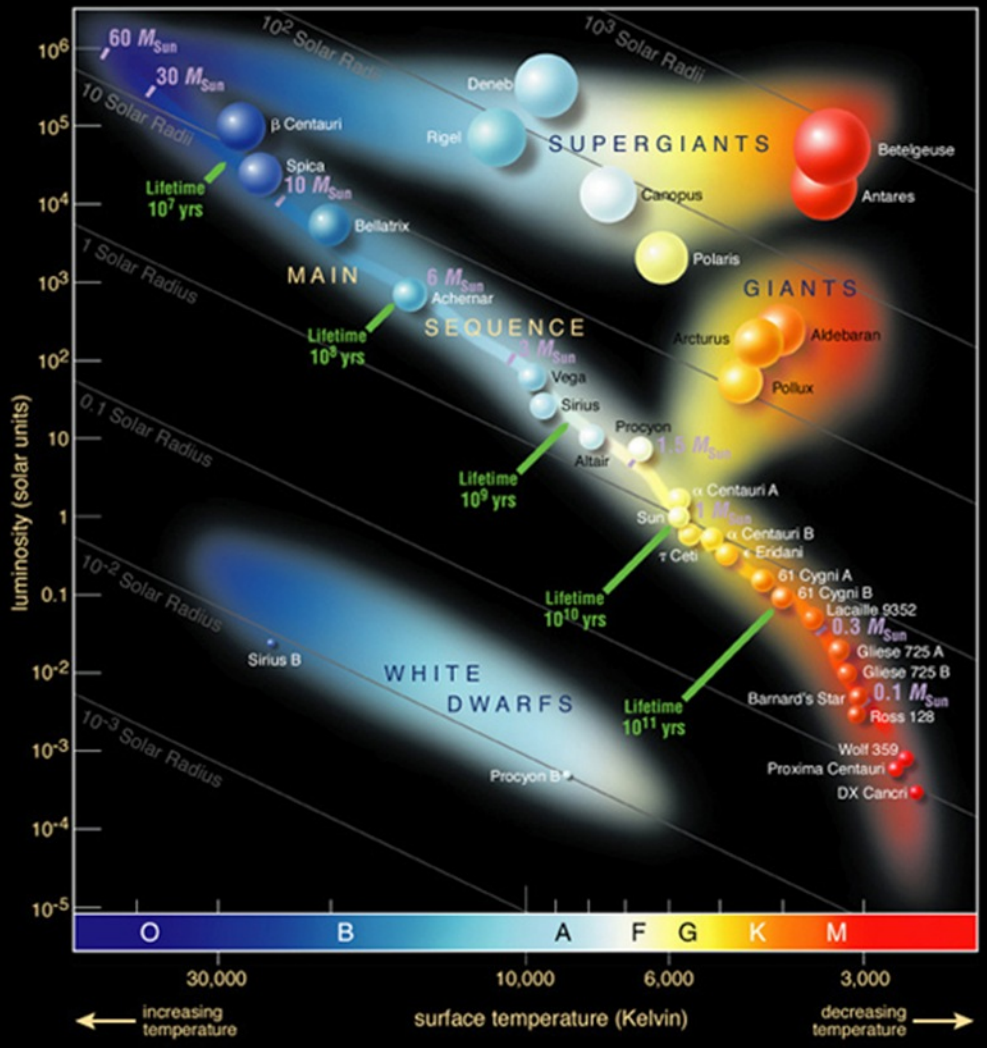
PLANETS

Saturn

STARS

Altair	mag=0.75	D=16.77 ly
Polar	mag=1.95	D=431 ly
Deneb	mag=1.1	D=3200 ly
Arcturus	mag=-0.05	D=37.6 ly
Vega	mag=0	D=25 ly

H-R diagram



Deneb

Arcturus

Aldebaran

Vega

Altair

Sun

Sirius B

STELLAR CLUSTERS

M92	glogular cluster	mag=6.3 D=27 kly
M39	open cluster	mag=5.5 D=1026ly
M7	open cluster	mag=3.3 D= 980 ly
M13	**globular cluster**	**mag=5.8 D=22 kly**
M22	globular cluster	mag=5.1 D=10 kly

Catálogo de Messier,1771

NEBULAE

M57	**planetary nebula (Lira, the ring nebula)**	**D=1800 ly**
M27	planetary nebula (Fox)	D=1360 ly

Catálogo de Messier,1771

OBSERVATIONAL ASTRONOMY @ AL 2025

Scientific goal

Position of the targets in the sky

Location of the observatory

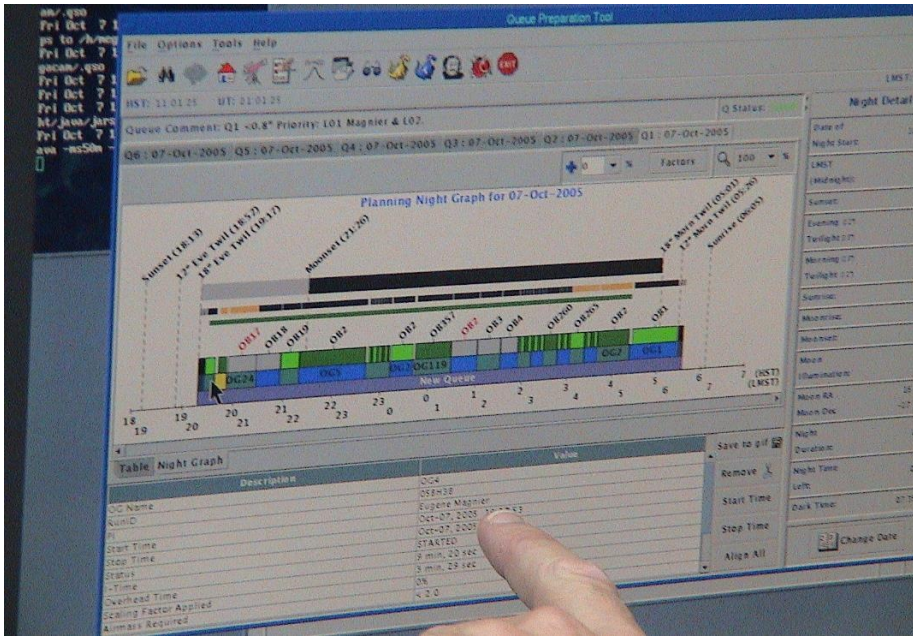
Characteristics of the telescopes

Observation without CCD

Observation with CCD

Planning of the observations: sequence of observations and exposure time

NEXT STEPS



Basic notions in observational astronomy

Prepare an observational poroposal

Prepare for the observations

Observations

Data reduction

Analysis and conclusions

Report

Proposals for the AO-CIGeoE

1. Make groups of max. 3 people and pick a scientific observing project that interests you

2. Find targets that are visible during the week of Sep. 11-26th 2025 from 20h-23h.

3. Write a small proposal (deadline: 11th Sep, 4pm:00) with abstract, scientific case and time justification

4. For time justification calculate exposure times and take into account overheads

Next lecture:
Basic notions in observational astronomy